Problem Statement

Data Set: Fire Department Calls for Service

URL: https://data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3

What are the requirements

1. Load the given data file and create a Spark data frame.
2. Use the Spark data frame to answer the following questions.
    1. How many distinct types of calls were made to the fire department?
    2. What are distinct types of calls made to the fire department?
    3. Find out all responses or delayed times greater than 5 mins?
    4. What were the most common call types?
    5. What zip codes accounted for the most common calls?
    6. What San Francisco neighborhoods are in the zip codes 94102 and 94103
    7. What was the sum of all calls, average, min, and max of the call response times?
    8. How many distinct years of data are in the CSV file?
    9. What week of the year in 2018 had the most fire calls?
    10. What neighborhoods in San Francisco had the worst response time in 2018?

Fire Department Calls for Service

data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3

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BETA Introducing our new data shaping and exploration experience: Filter, group, aggregate, and more! Try it now Learn more

Fire Department Calls for Service

Public Safety

View Data Visualize Export API ...

Fire Calls-For-Service includes all fire units responses to calls. Each record includes the call number, incident number, address, unit identifier, call type, and disposition. All relevant time intervals are also included. Because this dataset is based on responses, and since most calls involved multiple units, there are multiple records for each call number.

More

Updated
April 1, 2022

About this Dataset

Mute Dataset

Updated
April 1, 2022

Data Last Updated
April 1, 2022

Metadata Last Updated
April 1, 2022

Department Metrics

Publishing Department	Fire Department

Detailed Descriptive

Fire Department Calls for Service

data.sfgov.org/Public-Safety/Fire-Department-Calls-for-Service/nuek-vuh3

Columns in this Dataset

Column Name	Description	Type
Call Number		Plain Text
Unit ID		Plain Text
Incident Number		Plain Text
Call Type		Plain Text
Call Date		Date & Time
Watch Date		Date & Time
Received DtTm		Date & Time

Show All (34)

Table Preview

View Data | Create Visualization

Call ...	Unit ID	Incid...	Call T...	Call ...	Watc...	Recei...	Entry...	Disp...	Resp...	On S...	Tran...	Hos

What will you do?

1. How to load data into Spark
2. How to query data in Spark

Two Approaches

1. Using Spark SQL
2. Using Spark data frame API

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What will you learn?

1. Using Spark SQL and DDL statements
2. Spark Data Frame and Data Frame Reader API
3. Working with CSV Files
4. Data Frame Transformations and Actins
5. How to read Spark API Documentation

and more...

APACHE Spark™

What is Spark?

A Data Processing Platform

APACHE Spark™

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Database

Offers the following

1. Tables
2. SQL

Understanding Database Table

Table Data

Table Schema

Text	Text	Text	Date	Int	Double	Text
InvoiceNo	StockCode	Description	InvoiceDate	Quantity	UnitPrice	Country
565231	47504K	ENGLISH ROSE GARDEN SECATEURS	01-06-2022 9.26	1	3.29	United Kingdom
565231	47566	PARTY BUNTING	01-06-2022 9.26	2	10.79	United Kingdom
565231	51014A	FEATHER PEN,HOT PINK	01-06-2022 9.26	2	0.83	United Kingdom
565231	72760B	VINTAGE CREAM 3 BASKET CAKE STAND	01-06-2022 9.26	1	20.79	United Kingdom
565231	72807A	SET/3 ROSE CANDLE IN JEWELLED BOX	01-06-2022 9.26	1	8.29	United Kingdom
565231	72807C	SET/3 VANILLA SCENTED CANDLE IN BOX	01-06-2022 9.26	1	8.29	United Kingdom
565231	82551	LAUNDRY 15C METAL SIGN	01-06-2022 9.26	1	2.46	United Kingdom
565231	82567	AIRLINE LOUNGE,METAL SIGN	01-06-2022 9.26	1	1.63	United Kingdom

Logical Layer
Database Table

Physical Storage Layer
File in Hard Disk

Data File

Understanding Database Table

Metadata Layer
Table Schema

Meta Store

Logical Layer
Database Table

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Physical Storage Layer
File in Hard Disk

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Understanding Database Table

Metadata Layer
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Meta Store

SQL Engine

SELECT xyz
FROM INVOICES

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Physical Storage Layer
File in Hard Disk

Data File

Spark offers

1. Spark Database and SQL
2. Spark Data frame and API

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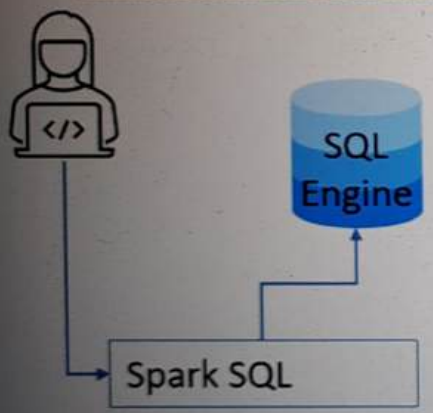
Understanding Spark Table

Metadata Layer
Table Schema



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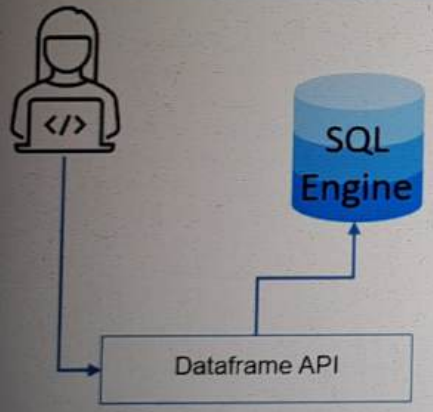
Physical Storage Layer
File in Distributed Storage

CSV, JSON, Parquet, AVRO, XML

Understanding Spark Data Frame

Metadata Layer
Data Frame Schema

Catalog



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Physical Storage Layer
File in Distributed Storage

CSV, JSON, Parquet, AVRO, XML

Data File

Spark Table Vs Data Frame

Spark Table

1. Tables store schema information in metadata store
2. Table and metadata are persistent objects and visible across applications
3. We create tables with a predefined table schema
4. Table supports SQL Expressions and does not support API

Spark Data Frame

1. Data Frame stores schema information in runtime Catalog
2. Data Frame and Catalog are runtime objects and live only during the application runtime. Data Frame is visible to your application only.
3. Data Frame supports schema-on-read
4. Data Frame offers APIs and does not support SQL expressions

→ Spark Table and a Data Frame are convertible objects

Data Engineering – ETL/ELT

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1. Extract
   - What : Read from source system
   - How: Ingestion tools / Spark Source System Connector

2. Load
   - What: Save to Data Lake
   - How: Ingestion Tools / Spark Source System Connector

3. Transform
   - What: Process / Prepare data as per your requirement
   - How: Spark Dataframe / Spark SQL
     1. Read from data lake: Spark Session
     2. Transform: Dataframe / SQL
     3. Save: Dataframe / SQL

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What is Spark Session

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An entry point to programming Spark with the Dataframe API

A Spark Session can be used to

- Connection to Cluster as Driver
- Create Dataframe
- Read Data Files
- Read Data Tables
- Execute SQL over Tables
- Access Runtime Configuration
- Access Spark Context

How to create Spark Session

- Spark Session Builder – Server Side
- Spark Connect Session Builder – Client Server
- Pre-created Spark Session – Platform Dependent

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Dataframe?

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A DataFrame is a two-dimensional labeled data structure

Spark Dataframe support a rich set of APIs to

1. Select
2. Filter
3. Join
4. Aggregate
5. Many other ...
6. Save

Spark Dataframe advantages for big data

1. Parallel processing
2. In-memory processing
3. Schema flexibility
4. Fault tolerance

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What is Spark Dataframe?

ScholarNest

Runtime in-memory data object (Data and Methods)

1. **Transformations**
2. **Actions**
3. DataFrameWriter
4. Auxiliary Methods